# Sample Midterm Exam 1

Econ 5030

## 1 Preferences

Consider an agent with preferences over $X = [0,1] \times [0,1]$ defined as follows: For all $x, x' \in X$,

$$x \succsim x' \quad \text{if and only if} \quad x_1 \geq x_1' \text{ and } x_2 \geq x_2'.$$

**1.1** Write the definition of complete preferences and clearly demonstrate that these preferences are (or are not) complete on $X$.

**1.2** Write the definition of transitive preferences and clearly demonstrate that these preferences are (or are not) transitive on $X$.

**1.3** Does a utility function exist that represents these preferences on $X$? Explain.

**1.4** Are these preferences continuous? Does your answer change your conclusion in 1.3? Explain.

## 2 Game Theory

### 2.1 Timing and Strategies

Take a simple two player game with each player having two actions. Player 1's set of actions is $\{a_1, b_1\}$. Player 2's set of actions is $\{c_2, d_2\}$. The players' payoffs over the four possible action combinations are as follows:

$$\begin{aligned} u_1(a_1, c_2) &= 40 & \qquad u_2(a_1, c_2) &= 20 \\ u_1(a_1, d_2) &= 10 & \qquad u_2(a_1, d_2) &= 10 \\ u_1(b_1, c_2) &= 25 & \qquad u_2(b_1, c_2) &= 30 \\ u_1(b_1, d_2) &= 20 & \qquad u_2(b_1, d_2) &= 15 \end{aligned}$$

Suppose this is a sequential choice game in which player 1 goes first, then player 2 observes player 1's choice and makes her choice.

**2.1** Write the Normal Form of the game.

**2.2** Construct the Game matrix.

**2.3** Find the set of strategies that remain after iterated deletion of strictly dominated strategies.

**2.4** Find the set of Nash Equilibrium.

**2.5** Write the Game Tree and find the set of Subgame perfect equilibrium.

### 2.2 Games with Different Preferences

There are two players and each player $i$ has two actions $\{R_i, Q_i\}$. The money each player $i$ receives for the different action pairs (in a game with player $j$) is described in the matrix below (*this is not the Game Matrix*):

| | | $P2$ | |
|---|---|---|---|
| | | $R_2$ | $Q_2$ |
| $P1$ | $R_1$ | $\$12, \$12$ | $\$2, \$16$ |
| | $Q_1$ | $\$16, \$2$ | $\$4, \$4$ |

The game is sequential. Player 1 picks her action first and then player 2 observes the action of player 1 and player 2 picks his action. We will consider 3 cases of this game.

*Case 1*: Both players only care about their own money with more money being strictly better than less.

**2.6** Create a normal form for this game. Write the game matrix and find the set of Nash Equilibrium. Draw the game tree and find the set of Subgame perfect equilibrium.

*Case 2*: Both players care about each other in a particular way. Each player only cares about the sum of the two players' money and strictly prefers more total money to less.

**2.7** Create a normal form for this game. Write the game matrix and find the set of Nash Equilibrium. Draw the game tree and find the set of Subgame perfect equilibrium.

*Case 3*: Player 1 only cares about her own money. Player 2 is one of two types, and Player 1 does not know which. With probability $\mu \in (0, 1)$, Player 2 is *selfish* (cares only about his own money). With probability $1 - \mu$, Player 2 is *inequity averse*: his payoff is his own money minus the absolute difference between the two players' money. Player 2 knows his own type.

**2.8** You will need to be a little creative to analyze this game. Create a game tree for this game and use a form of backward induction as your solution concept. For which values of $\mu$ does Player 1 choose $R_1$? Explain clearly the difficulties you faced in finding a way to analyze the game.

## 3 Conceptual Questions

**3.1** Explain clearly the idea of the Independence Axiom. Why is this the exact assumption needed for an expected utility form representation of preferences over lotteries?

**3.2** Define the Normal Form of a game. What are the three types of solution concepts we discussed in class for the Normal Form? How are they related?

**3.3** Define a Pareto efficient outcome. Is a Nash equilibrium always efficient? Is it always equitable? Use one of the games above in your answer.

**3.4** In a pure exchange (market) game with no production, what is the natural welfare measure? When are gains from trade maximized, and what gets in the way?

# 4 Cournot Oligopoly

Consider a Cournot duopoly. Firms $i = 1, 2$ simultaneously choose quantities $q_i \geq 0$ of a homogeneous good. Each firm has cost function $C(q_i) = q_i^2$. Market inverse demand is $P(Q) = 200 - Q$, where $Q = q_1 + q_2$.

**4.1** Write the Normal Form of this game.

**4.2** Write the profit of the arbitrary firm $i$. Find the best response function for firm $i$ and graph both best responses on the $q_1$–$q_2$ plane.

**4.3** Find the Cournot equilibrium quantities $q^c$, total quantity $Q^c$ and market price $p^c$.

**4.4** Find each firm's profit, the Consumer Surplus (CS), Producer Surplus (PS) and Total Surplus (TS).

**4.5** Find the efficient (total surplus maximizing) outcome and the Dead Weight Loss (DWL). Graph the market outcome on the $p$–$Q$ plane and label CS, PS and DWL.

**4.6** Explain why the Cournot outcome is not efficient.

**4.7** Now suppose there are $n$ identical firms, each with $C(q_i) = q_i^2$. Find the best response $b_i(Q_{-i})$, the symmetric Cournot quantity $q^c(n)$ and price $p^c(n)$. What happens as $n \to \infty$?